DE LA SALLE HOLY CROSS COLLEGE

Grade 10 November Examination | **Date: 11-11-2021**

Physical Science | **Total: 130 marks**

Paper 2: Chemistry | **Time: 2 hours**

Examiner: K. Tam | **Moderator: A. Kamedza**

INSTRUCTIONS AND INFORMATION

1. This question paper consists of 14 pages. The formula sheet and answer sheet are attached to the back of the paper. Detach the answer sheet and attach to the front of your folio paper.

2. This question paper consists of 7 questions. Answer ALL the questions on folio paper. **QUESTION 1 and QUESTION 3.3 to be done on the answer sheet.**

3. Write neatly and legibly.

4. Number the answers correctly according to the numbering system used in the question paper.

5. Leave ONE line open between two subsections, for example between QUESTION 2.1 and QUESTION 2.2.

6. You may use a non-programmable calculator.

7. You may use appropriate mathematical instruments.

8. Show ALL formulae and substitutions in ALL calculations.

9. Round off your final numerical answers to a minimum of TWO decimal places OR **three significant figures**.

Question 1: Multiple choice

The questions that follow are multiple choice questions. Four possible options are provided as answers. Each question has only one correct answer. Choose the correct answer and write the appropriate letter (A-D) In the empty block on the answer sheet on p. 13

1.1 Which of the following substances is a mixture?

A. Oxygen.
B. Table salt.
C. Calcium.
D. Air.

1.2 The correct formula of potassium dichromate is…

A. K_2CrO_4.
B. $K_2Cr_2O_7$.
C. K_2CrO_7.
D. $K_2Cr_2O_4$.

1.3 Polar covalent bonds are characterised by the…

A. transfer of electrons.
B. unequal sharing of electrons.
C. delocalised electrons.
D. equal sharing of electrons

1.4 The reaction of a covalent compound with water is known as…

A. dissociation.
B. dissolving.
C. ionisation.
D. melting.

1.5 The name of the group 18 elements is known as…

A. Halogens.
B. Diatomic elements.
C. Noble gases.
D. Inert gases.

1.6 Halogens gain electrons due to which of the following characteristics?

I) Have a high electron affinity.

II) Have a high ionisation energy.

III) Have a large atomic radius.

A. I only.
B. I and III.
C. II and III.
D. I and II.

1.7 Which ONE of the following groups of elements shows the correct trend of the atomic radii of elements?

A. F > Cl > Br > I
B. I > Br > Cl > F
C. Li < Be < B < N
D. Li > B > N > Be

1.8 Consider the reaction below:

$$2CO_{2(g)} + 2H_{2(g)} \rightarrow C_2H_{4(g)} + 2O_{2(g)}$$

At standard temperature and pressure identify the amount of hydrogen that would be required to react with 22,4 dm^3 of carbon dioxide.

A. 44,8 dm^3
B. 11,2 dm^3
C. 22,4 dm^3
D. 33,6 dm^3

1.9 The number of hydrogen ions that 1 mol of sulphuric acid makes is _____ $x10^{23}$ ions.

A. 6,02.
B. 3,02.
C. 12,04.
D. 24,08.

1.10 The empirical formula of a certain carbon compound is CH_2O. Identify which ONE of the following can be the molecular formula of the compound.

A. $C_3H_6O_3$.

B. C_3H_6O.

C. $C_2H_6O_2$.

D. $C_2H_6O_2$.

[10x2=20]

Question 2:

Chlorine is a non-metallic element with an atomic number of 17. It is an important industrial chemical as it is very reactive and it can exist as isotopes.

2.1 Define the term *isotope*. (2)

2.2 Natural chlorine consists of two isotopes Cl_{35} and Cl_{37}.

2.2.1 Write down the sp-notation for Cl_{37}. (2)

2.2.2 The relative atomic mass of chlorine is 35,5. Calculate the percentage of Cl_{35} in natural chlorine. (3)

2.3 Chlorine gas exists in nature as a diatomic molecule.

Write down the following:

2.3.1 The number of valence electrons each chlorine atom has. (1)

2.3.2 The specific type of bond between chlorine atoms. (2)

2.3.3 Draw the Lewis structure of the chlorine molecule. (1)

2.4 Calcium chloride is a product that forms when chlorine reacts with calcium.

2.4.1 Draw the Aufbau diagram of the calcium ion. (3)

2.4.2 Draw the Lewis structure of calcium chloride. (2)

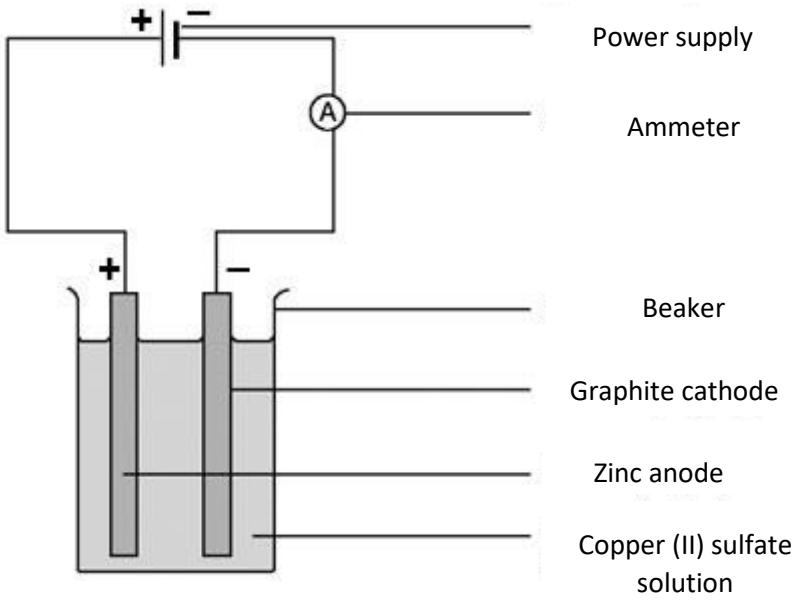
[16]

Question 3:

Acacia, Ambika and Chiara are given an experiment to perform. They are given the following materials and apparatus:

- Copper (II) sulphate solution of 1 mol.dm^{-3}.
- 5 x zinc electrodes.
- 5 x graphite electrodes of 80g.
- 5 x beakers.
- 5 x cells
- A measuring cylinder.
- A stopwatch.
- A thermometer.
- Scale.

The cells are set up as shown below:



The three girls were told that they are to measure the change in the mass of the graphite electrode which accumulates solid copper in 20 second intervals of the experiment.

The method is as follows.

1. Measure 50 ml of copper (II) sulphate in the measuring cylinder.
2. Pour 50 ml of copper (II) sulphate into each beaker.
3. Place the electrodes of zinc and graphite into each beaker simultaneously.
4. Start timing immediately as the electrodes are added.
5. Every 20 seconds remove a graphite electrode, rinse with distilled water and weigh the electrode on the scale.
6. Record your results in a table.

The results are as follows:

Time (s)	20	40	60	80	100
Mass of plate (g)	120	160	200	210	210

Answer the questions that follow:

3.1 Write the formula of copper (II) sulphate. (1)

3.2 Identify the following:

3.2.1 The dependant variable. (1)

3.2.2 The independent variable. (1)

3.2.3 TWO controlled variables. (2)

3.3 Draw a graph of the mass of the electrodes vs time on the **answer sheet provided on page 14**. (7)

3.4 Name the type of reaction occurring in the beakers. (2)

3.5 Using the graph, calculate the average rate at which copper forms on the zinc plates in grams per second. (3)

3.6 Explain why there was no difference between the 80 second and 100 second zinc plate. (1)

[18]

Question 4:

4.1 Sulfuric acid, H_2SO_4, is a strong mineral acid which is incredibly corrosive and dangerous. It strongly reacts with water in the process shown below:

$$H_2SO_4 + 2H_2O \rightarrow H_3O^+ + SO_4^{2-}$$

Name the above reaction. (2)

4.2 A few drops of bromothymol blue are added to a beaker of the sulfuric acid solution and solid magnesium is added to the beaker until a colour change occurs.

4.2.1 Give the balanced chemical equation for the reaction between magnesium and H_2SO_4 (2)

4.2.2 Define the term *indicator.* (2)

4.2.3 Describe a method which you would use to identify or test for the gas that is formed. (2)

4.3 Potassium hydroxide, KOH, reacts fully with sulfuric acid. The balanced chemical reaction is given below.

$$2KOH_{(aq)} + H_2SO_{4(aq)} \rightarrow K_2SO_{4(aq)} + 2H_2O_{(l)}$$

4.3.1 Identify the type of reaction between KOH and H_2SO_4. (1)

4.3.2 200 ml of KOH with a concentration of 0,05 mol.dm^{-3}, reacts with 50 ml of H_2SO_4. Calculate the concentration of H_2SO_4. (5)

4.3.3 Name the metal salt produced in the above reaction. (1)

4.4 In a sample of acid it is determined that acid is made up of 3,2 % hydrogen, 19,4 % carbon and 77,4 % oxygen. Determine the empirical formula of this acid. (3)

[18]

Question 5:

Electronegativity is a property which is used to determine polarity of molecules and to predict the characteristics of different compounds interacting in chemical reactions.

5.1.1 Define *electronegativity.* (2)

5.1.2 Calculate the electronegativity difference between hydrogen and chlorine in HCl. (2)

5.1.3 The molecules of HCl and Cl_2 are both covalently bonded but HCl is polar whereas Cl_2 is not. Explain, using electronegativities, why this is the case. (4)

5.2 Study the table below of first ionisation energies and electron affinities of the period three elements.

ELEMENT	FIRST IONISATION ENERGY ($kJ \cdot mol^{-1}$)	ELECTRON AFFINITY ($kJ \cdot mol^{-1}$)
Sodium	496	53
Magnesium	738	0
Aluminium	578	44
Silicon	786	134
Phosphorous	1 012	72
Sulphur	1 000	200
Chlorine	1 251	349
Argon	1 521	0

5.2.1 Define *electron affinity.* (2)

5.2.2 Identify the trend in the first ionisation energies given. (2)

5.2.3 How will the SECOND ionisation energy of sodium compare to magnesium? Write only HIGHER THAN, LOWER THAN or EQUAL TO. (1)

5.2.4 Explain your answer to **QUESTION 5.2.3.** (2)

[15]

Question 6:

6.1 Mercury (II) oxide is a red/orange powder which can go through a reaction where mercury metal and oxygen gas is formed.

6.1.1 Write the formula of mercury (II) oxide. (1)

6.1.2 State the type of reaction that occurs with mercury (II) oxide. (2)

6.1.3 Write a balanced equation for the chemical reaction of mercury (II) oxide mentioned above. (3)

6.2 Study the balanced chemical equation of the reaction between sodium carbonate (Na_2CO_3) and hydrochloric acid (HCl) and answer the questions that follow.

$$Na_2CO_3 + 2HCl \rightarrow 2NaCl + CO_2 + H_2O$$

6.2.1 Identify the type of reaction above. (1)

6.2.2 Identify a test for CO_2 and state what indicates the presence of the gas. (2)

6.3 In the reaction in **QUESTION 6.2**, 10,6g of sodium carbonate reacts in excess hydrochloric acid. Calculate the following:

6.3.1 The moles of sodium carbonate that reacted. (3)

6.3.2 The mass of H_2O produced. (3)

6.3.3 The volume of CO_2 produced at STP. (4)

6.4 A sample of hydrated sodium carbonate $Na_2CO_3.\mathbf{x}H_2O$, was heated until no further change in mass was recorded. On heating all the water of crystallisation was evaporated.

The results of the experiment are recorded in the table below.

Item	Mass (g)
Crucible	50,4
Crucible and $Na_2CO_3.xH_2O$	64,7
Crucible and Na_2CO_3	55,7

Calculate the following.

6.4.1 The amount in mol of H_2O lost due to heating. (3)

6.4.2 The molecular formula of hydrated sodium carbonate. (6)

[28]

Question 7:

The grade 10 learners can use a specific type of reaction to identify ions in solutions. These ions are sulphates, halides and carbonates.

Learners are taken to the laboratories and given three conical flasks with samples of water, *sample A, sample B and sample C*. Each sample has a different ion present.

Their task is to identify the ions present in each flask.

7.1 Name the type of reaction that produces an *insoluble salt*. (2)

7.2 In *sample A,* barium chloride ($BaCl_2$) reacted to form an *insoluble salt*.

7.2.1 Name the two ions that could be present in *sample A*. (2)

7.2.2 State the colour of the insoluble salt. (1)

7.2.3 How would you distinguish between the two ions identified in QUESTION 7.2.1? (2)

7.3 *Sample B* and *Sample C* have been determined to have magnesium salts. They both form an insoluble salt in the presence of silver nitrate. *Sample B* is cream coloured, and *Sample C* is yellow coloured.

7.3.1 Give the name and chemical symbol of the ion in *Sample B*. (2)

7.3.2 Give the name and chemical symbol of the ion in *Sample C.* (2)

7.3.3 Write the balanced equation for the reaction in S*ample B.* (4)

[15]

Total: [130]

END

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 2,1 **H** 1			Atomic number (Z)			1 2,1 **H** 1		Electronegativity									2 **He** 4
2	3 1,0 **Li** 7	4 1,5 **Be** 9					Relative atomic mass						5 2,0 **B** 10,8	6 2,5 **C** 12	7 3,0 **N** 14	8 3,5 **O** 16	9 4,0 **F** 19	10 **Ne** 20
3	11 0,9 **Na** 23	12 1,2 **Mg** 24,3											13 1,5 **Aℓ** 27	14 1,8 **Si** 28	15 2,1 **P** 31	16 2,5 **S** 32	17 3,0 **Cℓ** 35,5	18 **Ar** 40
4	19 0,8 **K** 39	20 1,0 **Ca** 40	21 1,3 **Sc** 45	22 1,5 **Ti** 48	23 1,6 **V** 51	24 1,6 **Cr** 52	25 1,5 **Mn** 55	26 1,8 **Fe** 56	27 1,8 **Co** 59	28 1,8 **Ni** 59	29 1,9 **Cu** 63,5	30 1,6 **Zn** 65,4	31 1,6 **Ga** 70	32 1,8 **Ge** 72,6	33 2,0 **As** 75	34 2,4 **Se** 79	35 2,8 **Br** 80	36 **Kr** 84
5	37 0,8 **Rb** 85,5	38 1,0 **Sr** 88	39 1,2 **Y** 89	40 1,4 **Zr** 91	41 1,6 **Nb** 93	42 1,8 **Mo** 96	43 1,9 **Tc** 99	44 2,2 **Ru** 101	45 2,2 **Rh** 103	46 2,2 **Pd** 106	47 1,9 **Ag** 108	48 1,7 **Cd** 112	49 1,7 **In** 115	50 1,8 **Sn** 119	51 1,9 **Sb** 121	52 2,1 **Te** 128	53 2,5 **I** 127	54 **Xe** 131
6	55 **Cs** 133	56 **Ba** 137,3		72 **Hf** 178,5	73 **Ta** 181	74 **W** 184	75 **Re** 186	76 **Os** 190	77 **Ir** 192	78 **Pt** 195	79 **Au** 197	80 **Hg** 200,6	81 **Tℓ** 204,4	82 **Pb** 207	83 **Bi** 209	84 **Po** –	85 **At** –	86 **Rn** –
7	87 **Fr**	88 **Ra**																

57 **La**	58 **Ce**	59 **Pr**	60 **Nd**	61 **Pm**	62 **Sm**	63 **Eu**	64 **Gd**	65 **Tb**	66 **Dy**	67 **Ho**	68 **Er**	69 **Tm**	70 **Yb**	71 **Lu**
89 **Ac**	90 **Th**	91 **Pa**	92 **U**	93 **Np**	94 **Pu**	95 **Am**	96 **Cm**	97 **Bk**	98 **Cf**	99 **Es**	100 **Fm**	101 **Md**	102 **No**	103 **Lw**

TABLE 1 PHYSICAL CONSTANTS

NAME	SYMBOL	VALUE
Magnitude of charge on electron	e	$1{,}6 \times 10^{-19}$ C
Mass of an electron	m_e	$9{,}1 \times 10^{-31}$ kg
Standard pressure	p^θ	$1{,}01 \times 10^5$ Pa
Molar gas volume at STP	V_m	22,4 $dm^3 \cdot mol^{-1}$
Standard temperature	T^θ	273 K
Avogadro's constant	N_A	$6{,}02 \times 10^{23}$ mol^{-1}
Faraday's constant	F	96 500 $C \cdot mol^{-1}$

TABLE 2 FORMULA

$n = \frac{m}{M}$	$n = \frac{N}{N_A}$
$n = \frac{V}{V_m}$	$c = \frac{n}{V}$ or $c = \frac{m}{MV}$

GRADE 10 PHYSICAL SCIENCE PAPER 2 ANSWER SHEET

NAME: _____ **DATE:** _____

QUESTION	OUT OF	MARK
1	20	
2	16	
3	18	
4	18	
5	15	
6	28	
7	15	
TOTAL	130	

QUESTION 1

1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10

QUESTION 3.3

